

Multiple Choice

1. **Answer:** A

Options: A) SE Asia; B) Russia; C) Africa and S. America; D) N. America

Note: Arenaviruses are predominantly found in Southeast Asia due to the region's ecological conditions and the presence of specific rodent hosts that facilitate the transmission of these viruses to humans.

2. **Answer:** A

Options: A) Height; B) Bone structure; C) Immune Function; D) Diet and exercise

Note: BMI is considered a better measure of health than weight alone because it incorporates height, allowing for a more accurate assessment of body composition relative to an individual's stature, which helps differentiate between fat mass and lean mass.

3. **Answer:** B

Options: A) Alzheimer's; B) Korsakoff's; C) Creutzfeldt-Jakob; D) Lewy body

Note: Korsakoff's dementia is specifically linked to chronic thiamine (vitamin B₁) deficiency, commonly associated with alcohol misuse, leading to memory disturbances and cognitive impairments.

4. **Answer:** C

Options: A) will produce a high voice because of decreased testosterone production.; B) will produce a high voice because of the pain that accompanies the procedure.; C) will not produce a high voice because the larynx has already grown.; D) will not produce a high voice because castration does not affect testosterone production.

Note: Castration in adulthood will not produce a high voice because by that stage, the larynx has already fully developed and matured, resulting in a deeper voice that cannot be altered by hormonal changes.

5. **Answer:** D

Options: A) mosaicism.; B) diploidy.; C) aneuploidy.; D) chimaerism.

Note: Chimaerism refers to the condition in which an individual possesses two or more genetically distinct cell lines derived from different zygotes, resulting from the fusion of multiple embryos during early development.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Sexual intercourse typically involves physical maturity and consent, which are not common characteristics of preadolescents, making it an uncommon form of homosexual activity during this developmental stage.

7. **Answer:** False

Options: A) True; B) False

Note: Ebbinghaus is primarily known for his research on memory and forgetting using experimental methods, particularly his studies on nonsense syllables, rather than conducting a longitudinal study on memory retrieval difficulties as people age.

8. **Answer:** True

Options: A) True; B) False

Note: Patients with pernicious anemia have an inability to absorb vitamin B-12 due to a lack of intrinsic factor, which is why they are treated with injections of vitamin B-12 to bypass the gastrointestinal absorption issue and correct the deficiency.

9. **Answer:** True

Options: A) True; B) False

Note: Recombination during meiosis occurs in Prophase I because this is the stage where homologous chromosomes pair up and exchange segments of genetic material through a process known as crossing over, leading to genetic diversity in gametes.

10. **Answer:** False

Options: A) True; B) False

Note: Psychological definitions of sexual behavior often take into account specific sexual preferences and behaviors, categorizing individuals based on their sexual orientations, identities, and practices.

Long Form Response

11. **Answer:** The papilloma family of viruses, particularly human papillomavirus (HPV), can lead to cancer through the integration of their viral genomes into the host cellular DNA. This integration disrupts normal cellular functions and can activate oncogenes or inactivate tumor suppressor genes. For instance, the viral oncoproteins E₆ and E₇ interfere with the host's p 53 and Rb tumor suppressor pathways, which are critical for regulating the cell cycle and preventing uncontrolled cell proliferation. As a result, the infected cells may undergo malignant transformation, ultimately leading to the development of cancers such as cervical cancer. Thus, the ability of papillomaviruses to integrate their genetic material into the host genome is a key mechanism by which they promote oncogenesis.

Note: correct because it accurately describes how the integration of HPV genomes into host DNA disrupts cellular regulation by inactivating tumor suppressor genes like p 53 and Rb, leading to uncontrolled cell proliferation and cancer development.

12. **Answer:** Cocaine abuse during pregnancy has significant health implications that affect maternal health, fetal development, and the potential long-term outcomes for the child. Maternal use of cocaine can lead to increased risks of complications such as placental abruption, preterm labor, and maternal cardiovascular issues. For the fetus, exposure to cocaine can result in developmental problems, low birth weight,

and neurobehavioral issues, as the drug can interfere with essential brain development. Long-term outcomes for children exposed to cocaine in utero may include cognitive deficits, behavioral problems, and an increased risk of substance abuse later in life. Overall, the ramifications of cocaine abuse during pregnancy underscore the critical need for targeted interventions and support for affected mothers.

Note: correct because it comprehensively addresses the multifaceted health risks associated with cocaine abuse during pregnancy, detailing its adverse effects on maternal health, fetal development, and the potential for long-term developmental issues in children.